

UFUG 2106 : Discrete Mathematics

Chapter 01: Speaking Mathematically



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Recommended readings: Pages 1-13; 15-21; 24-34

ACK: Part of the PPTs are built from the PPT slides from Instructor's Companion Website of the textbook.

Outline

1. Variables
2. Sets
3. Relations and Functions
4. Graphs

Learning outcomes:

- Understand the basic concepts of variables, statements, sets, relations, functions, and graphs
- Be able to rewrite statements in human language using variables
- Be able to find cartesian products of sets
- Be able to use set theory to define relations and functions
- Be able to use graphs to model real-world applications

This chapter is a warm-up for the course. It introduces the basic concepts and terminologies that are fundamental to discrete mathematics.

We will discuss more details of sets, relations, functions, graphs in later chapters.

Variables

- In mathematics, a **variable** is a symbol (e.g., x , y , z) that holds a place for some numbers. E.g., $A = \pi r^2$.
- In computer science, a **variable** is an abstract storage location paired with an associated symbolic name, which contains some quantity of data. E.g., `int a = 10;`
- From the textbook: a **variable** is sometimes thought of as a mathematical “**John Doe**” because you can use it as a **placeholder** when you want to talk about something but either:
 - (1) you imagine that it has one or more values but you don’t know what they are, or
 - (2) you want whatever you say about it to be equally true for all elements in a given set, and so you don’t want to be restricted to considering only a particular, concrete value for it.

Use ChatGPT or other AI tools to find out what “John Doe” refers to.

Case I

To illustrate the first use, consider asking:

Is there a number with the following property: doubling **it** and adding 3 gives the same result as squaring **it**?

In this sentence you can introduce a **variable** to replace the potentially ambiguous word “**it**”:

Is there a number x with the property that $2x + 3 = x^2$?

The advantage of using a **variable**: it allows you to give a temporary name to what you are seeking so that you can perform concrete computations with it to help discover its possible values.

Case 2

To illustrate the second use of variables, consider the statement:

No matter what number might be chosen, if **it** is greater than 2, then **its square** is greater than 4.

In this case, introducing a variable to give a temporary name to the (arbitrary) number you might choose enables you to maintain the generality of the statement, and replacing all instances of the word “**it**” by the name of the variable ensures that possible ambiguity is avoided

No matter what number n might be chosen, if n is greater than 2, then n^2 is greater than 4.

From human language → mathematical language

Example 1.1.1 – *Writing Sentences Using Variables*

Use variables to rewrite the following sentences more formally.

- a. Are there two numbers with the property that the sum of their squares equals the square of their sum?
- b. Given any real number, its square is nonnegative.

Some Important Kinds of Mathematical Statements

Three of the most important kinds of sentences in mathematics are **universal statements**, **conditional statements**, and **existential statements**:

A **universal statement** says that a certain property is true for all elements in a set. (For example: *All positive numbers are greater than zero.*)

A **conditional statement** says that if one thing is true then some other thing also has to be true. (For example: *If 378 is divisible by 18, then 378 is divisible by 6.*)

Given a property that may or may not be true, an **existential statement** says that there is at least one thing for which the property is true. (For example: *There is a prime number that is even.*)

Remark: we will formally discuss **set** very soon.

Universal Conditional Statements

Universal statements contain some variation of the words “all / for every” and conditional statements contain versions of the words “if-then.”

A **universal conditional statement** is a statement that is both universal and conditional. Here is an example:

For every animal a , if a is a dog, then a is a mammal.

One of the most important facts about **universal conditional statements** is that they can be rewritten in ways that make them appear to be purely universal or purely conditional.

Universal Conditional Statements

For example, the previous statement can be rewritten in a way that makes its conditional nature explicit but its universal nature implicit:

If a is a dog, then a is a mammal.

Or: If an animal is a dog, then the animal is a mammal.

The statement can also be expressed so as to make its universal nature explicit and its conditional nature implicit:

For every dog a , a is a mammal.

Or: All dogs are mammals.

Example 1.1.2 – *Rewriting a Universal Conditional Statement*

Fill in the blanks to rewrite the following statement:

For every real number x , if x is nonzero then x^2 is positive.

- a. If a real number is nonzero, then its square _____.
- b. For every nonzero real number x , _____.
- c. If x _____, then _____.
- d. The square of any nonzero real number is _____.
- e. All nonzero real numbers have _____.

Universal Existential Statements

A **universal existential statement** is a statement that:

(1) is **universal** because its first part says that a certain property is true for all objects of a given type, and

(2) is **existential** because its second part asserts the existence of something.
For example:

Every real number has an additive inverse.

In this statement the property “has an additive inverse” applies universally to all real numbers. “Has an additive inverse” asserts the existence of something—an additive inverse—for each real number.

Universal Existential Statements

However, the nature of the additive inverse depends on the real number; different real numbers have different additive inverses. Knowing that an additive inverse is a real number, you can rewrite this statement in several ways, some less formal and some more formal:

All real numbers have additive inverses.

Or: For every real number r , there is an additive inverse for r .

Or: For every real number r , there is a real number s such that s is an additive inverse for r .

Example 1.1.3 – *Rewriting a Universal Existential Statement*

Fill in the blanks to rewrite the following statement:

Every pot has a lid.



- a. All pots _____.
- b. For every pot P , there is _____.
- c. For every pot P , there is a lid L such that _____.

Existential Universal Statements

An **existential universal statement** is a statement that:

- (1) is existential because its first part asserts that a certain object exists, and
- (2) is universal because its second part says that the object satisfies a certain property for all things of a certain kind. For example:

There is a positive integer that is less than or equal to every positive integer.

This statement is true because the number one is a positive integer, and it satisfies the property of being less than or equal to every positive integer.

Existential Universal Statements

We can rewrite the statement in several ways, some less formal and some more formal:

Some positive integer is less than or equal to every positive integer.

Or: There is a positive integer m that is less than or equal to every positive integer.

Or: There is a positive integer m such that every positive integer is greater than or equal to m .

Or: There is a positive integer m with the property that for every positive integer n , $m \leq n$.

Universal Existential and Conditional Statements

Some of the most important mathematical concepts, such as the definition of limit of a sequence, can only be defined using phrases that are universal, existential, and conditional, and they require the use of all three phrases “for every,” “there is,” and “if-then.”

For example, if $a_1, a_2, a_3, \dots$ is a sequence of real numbers, saying that

the limit of a_n as n approaches infinity is L

means that

for every positive real number ε , **there is** an integer N such that **for every** integer n , **if** $n > N$ **then** $-\varepsilon < a_n - L < \varepsilon$.

Outline

1. Variables

2. Sets

3. Relations and Functions

4. Graphs

The Language of Sets

Use of the word **set** as a formal mathematical term was introduced in 1879 by Georg Cantor (1845–1918).

For most mathematical purposes, a set is simply a collection of elements.

Set-Roster Notation

If S is a set, the notation $x \in S$ means that x is an element of S . The notation $x \notin S$ means that x is not an element of S . A set may be specified using the **set-roster notation** by writing all of its elements between braces. For example, $\{1, 2, 3\}$ denotes the set whose elements are 1, 2, and 3. A variation of the notation is sometimes used to describe a very large set, as when we write $\{1, 2, 3, \dots, 100\}$ to refer to the set of all integers from 1 to 100. A similar notation can also describe an infinite set, as when we write $\{1, 2, 3, \dots\}$ to refer to the set of all positive integers. (The symbol $\dots$ is called an **ellipsis** and is read “and so forth.”)

Set Theory

Set theory is the branch of mathematical logic that studies **sets**, which can be informally described as collections of objects.

The **axiom of extension** says that a set is completely determined by what its elements are — not the order in which they might be listed or the fact that some elements might be listed more than once.

- Two sets A and B are equal if and only if A and B have the same members.

Example 1.2.1 – *Using the Set-Roster Notation*

- a. Let $A = \{1, 2, 3\}$, $B = \{3, 1, 2\}$, and $C = \{1, 1, 2, 3, 3, 3\}$. What are the elements of A , B , and C ? How are A , B , and C related?
- b. Is $\{0\} = 0$?
- c. How many elements are in the set $\{1, \{1\}\}$?
- d. For each nonnegative integer n , let $U_n = \{n, -n\}$. Find U_1 , U_2 , and U_0 .

The Language of Sets

Certain sets of numbers are so frequently referred to that they are given special symbolic names. These are summarized in the following table.

Symbol	Set
R	the set of all real numbers
Z	the set of all integers
Q	the set of all rational numbers, or quotients of integers

The Language of Sets

Addition of a superscript + or – or the letters *nonneg* indicates that only the positive or negative or nonnegative elements of the set, respectively, are to be included. Thus

$\mathbf{R}^+$ denotes the set of positive real numbers, and

$\mathbf{Z}^{nonneg}$ refers to the set of nonnegative integers: 0, 1, 2, 3, 4, and so forth.

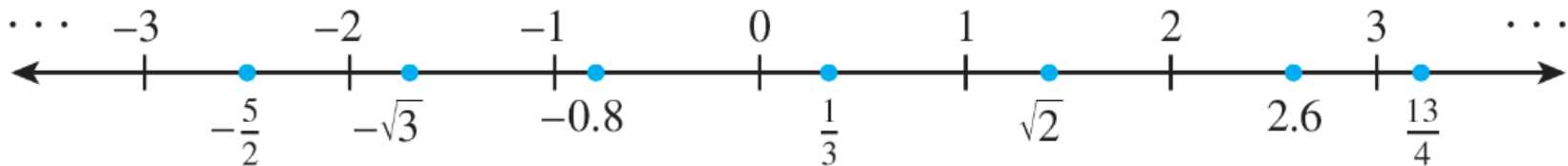
Some authors refer to the set of nonnegative integers as the set of **natural numbers** and denote it as **N**.

The Language of Sets

The set of real numbers is usually pictured as the set of all points on a line. The number 0 corresponds to a middle point, called the **origin**.

A unit of distance is marked off, and each point to the right of the origin corresponds to a positive real number found by computing its distance from the origin.

The set of real numbers is therefore divided into three parts: the set of positive real numbers, the set of negative real numbers, and the number 0.



The Language of Sets

Set-Builder Notation

Let S denote a set and let $P(x)$ be a property that elements of S may or may not satisfy. We may define a new set to be **the set of all elements x in S such that $P(x)$ is true**. We denote this set as follows:

$$\{x \in S \mid P(x)\}$$

the set of all such that

Example 1.2.2 – *Using the Set-Builder Notation*

Given that $\mathbf{R}$ denotes the set of all real numbers, $\mathbf{Z}$ the set of all integers, and $\mathbf{Z}^+$ the set of all positive integers, describe each of the following sets.

a. $\{x \in \mathbf{R} \mid -2 < x < 5\}$

b. $\{x \in \mathbf{Z} \mid -2 < x < 5\}$

c. $\{x \in \mathbf{Z}^+ \mid -2 < x < 5\}$

Subsets and Proper Subset

A basic **relation** between sets is that of subset.

Definition

If A and B are sets, then A is called a **subset** of B , written $A \subseteq B$, if, and only if, every element of A is also an element of B .

Symbolically:

$A \subseteq B$ means that for every element x , if $x \in A$ then $x \in B$.

The phrases *A is contained in B* and *B contains A* are alternative ways of saying that A is a subset of B .

For a set A not to be a subset of a set B means that there is at least one element of A that is not an element of B . Symbolically:

$A \not\subseteq B$ means that there is at least one element x such that $x \in A$ and $x \notin B$.

Definition

Let A and B be sets. A is a **proper subset** of B if, and only if, every element of A is in B but there is at least one element of B that is not in A .

Example 1.2.3 – Subsets

Let $A = \mathbf{Z}^+$, $B = \{n \in \mathbf{Z} \mid 0 \leq n \leq 100\}$,

and $C = \{100, 200, 300, 400, 500\}$. Evaluate the truth and falsity of each of the following statements.

a. $B \subseteq A$

b. C is a proper subset of A

c. C and B have at least one element in common

d. $C \subseteq B$

e. $C \subseteq C$

Example 1.2.4 – *Distinction between $\in$ and $\subseteq$*

Which of the following are true statements?

a. $2 \in \{1, 2, 3\}$

b. $\{2\} \in \{1, 2, 3\}$

c. $2 \subseteq \{1, 2, 3\}$

d. $\{2\} \subseteq \{1, 2, 3\}$

e. $\{2\} \subseteq \{\{1\}, \{2\}\}$

f. $\{2\} \in \{\{1\}, \{2\}\}$

Ordered Pair

Notation

Given elements a and b , the symbol (a, b) denotes the **ordered pair** consisting of a and b together with the specification that a is the first element of the pair and b is the second element. Two ordered pairs (a, b) and (c, d) are equal if, and only if, $a = c$ and $b = d$. Symbolically:

$$(a, b) = (c, d) \quad \text{means that} \quad a = c \text{ and } b = d.$$

Example 1.2.5 – *Ordered Pairs*

a. Is $(1, 2) = (2, 1)$?

b. Is $(3, \frac{5}{10}) = (\sqrt{9}, \frac{1}{2})$?

c. What is the first element of $(1, 1)$?

Ordered n-tuple

The notation for an **ordered n-tuple** generalizes the notation for an ordered pair to a set with any finite number of elements. It also takes both order and multiplicity into account.

Definition

Let n be a positive integer and let $x_1, x_2, \dots, x_n$ be (not necessarily distinct) elements. The **ordered n -tuple**, $(x_1, x_2, \dots, x_n)$, consists of $x_1, x_2, \dots, x_n$ together with the ordering: first x_1 , then x_2 , and so forth up to x_n . An ordered 2-tuple is called an **ordered pair**, and an ordered 3-tuple is called an **ordered triple**.

Two ordered n -tuples $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ are **equal** if, and only if, $x_1 = y_1, x_2 = y_2, \dots$, and $x_n = y_n$.

Symbolically:

$$(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n) \iff x_1 = y_1, x_2 = y_2, \dots, x_n = y_n.$$

Cartesian Products

Definition

Given sets $A_1, A_2, \dots, A_n$, the **Cartesian product** of $A_1, A_2, \dots, A_n$, denoted $A_1 \times A_2 \times \dots \times A_n$, is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ where $a_1 \in A_1$, $a_2 \in A_2, \dots, a_n \in A_n$.

Symbolically:

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n\}.$$

In particular,

$$A_1 \times A_2 = \{(a_1, a_2) \mid a_1 \in A_1 \text{ and } a_2 \in A_2\}$$

is the Cartesian product of A_1 and A_2 .

Example 1.2.7 – Cartesian Products

Let $A = \{x, y\}$, $B = \{1, 2, 3\}$, and $C = \{a, b\}$.

- a. Find $A \times B$.
- b. Find $B \times A$.
- c. Find $A \times A$.
- d. How many elements are in $A \times B$, $B \times A$, and $A \times A$?
- e. Find $(A \times B) \times C$.
- f. Find $A \times B \times C$.
- g. Let $\mathbf{R}$ denote the set of all real numbers. Describe $\mathbf{R} \times \mathbf{R}$.

Strings

Another notation, which is important in both mathematics and computer science, denotes objects called **strings**.

Definition

Let n be a positive integer. Given a finite set A , a **string of length n over A** is an ordered n -tuple of elements of A written without parentheses or commas. The elements of A are called the **characters** of the string. The **null string** over A is defined to be the “string” with no characters. It is often denoted λ and is said to have length 0. If $A = \{0, 1\}$, then a string over A is called a **bit string**.

Example 1.2.8 – *Strings*

Let $A = \{a, b\}$. List all the strings of length 3 over A with at least two characters that are the same.

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The Language of Relations and Functions

Example of **relation**: $S = \{\text{Jack, Rose, Zoe}\}$, $C = \{\text{UFUG1601, UFUG2106}\}$

A relation between **student** and **course** can be “a student registered a course”.

How do we represent the concept of relations mathematically?

Definition

Let A and B be sets. A **relation R from A to B** is a subset of $A \times B$. Given an ordered pair (x, y) in $A \times B$, **x is related to y by R** , written $x R y$, if, and only if, (x, y) is in R . The set A is called the **domain** of R and the set B is called its **co-domain**.

The notation for a relation R may be written symbolically as follows:

$x R y$ means that $(x, y) \in R$.

The notation $x \not R y$ means that x is not related to y by R :

$x \not R y$ means that $(x, y) \notin R$.

If Jack and Rose registered UFUG2106, and Rose and Zoe registered UFUG1601, then $R = \{(\text{Jack, UFUG2106}), (\text{Rose, UFUG2106}), (\text{Rose, UFUG1601}), (\text{Zoe, UFUG1601})\}$

Example 1.3.1 – *A Relation as a Subset*

Let $A = \{1, 2\}$ and $B = \{1, 2, 3\}$ and define a relation R from A to B as follows: Given any $(x, y) \in A \times B$,

$$(x, y) \in R \quad \text{means that} \quad \frac{x - y}{2} \text{ is an integer.}$$

- a. State explicitly which ordered pairs are in $A \times B$ and which are in R .
- b. Is $1 R 3$? Is $2 R 3$? Is $2 R 2$?
- c. What are the domain and co-domain of R ?

Arrow Diagram of a Relation

Suppose R is a relation from a set A to a set B . The arrow diagram for R is obtained as follows:

1. Represent the elements of A as points in one region and the elements of B as points in another region.
2. For each x in A and y in B , draw an arrow from x to y if, and only if, x is related to y by R . Symbolically:

Draw an arrow from x to y

if, and only if, $x R y$

if, and only if, $(x, y) \in R$.

Example 1.3.3 – Arrow Diagrams of Relations

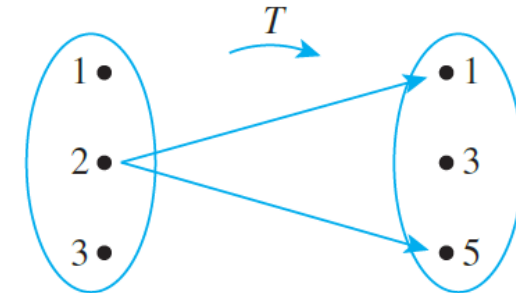
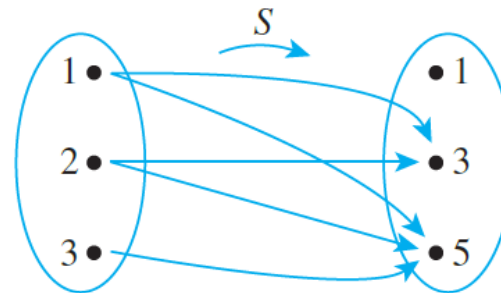
Let $A = \{1, 2, 3\}$ and $B = \{1, 3, 5\}$ and define relations S and T from A to B as follows:

For every $(x, y) \in A \times B$,

$(x, y) \in S$ means that $x < y$ S is a “less than” relation.

$T = \{(2, 1), (2, 5)\}$.

Draw arrow diagrams for S and T .



Functions

Definition

A **function F from a set A to a set B** is a relation with domain A and co-domain B that satisfies the following two properties:

1. For every element x in A , there is an element y in B such that $(x, y) \in F$.
2. For all elements x in A and y and z in B ,

if $(x, y) \in F$ and $(x, z) \in F$, then $y = z$.

Properties (1) and (2) can be stated less formally as follows: A relation F from A to B is a function if, and only if:

1. Every element of A is the first element of an ordered pair of F .
2. No two distinct ordered pairs in F have the same first element.

Function Notation

If A and B are sets and F is a function from A to B , then given any element x in A , the unique element in B that is related to x by F is denoted $F(x)$, which is read “ **F of x .**”

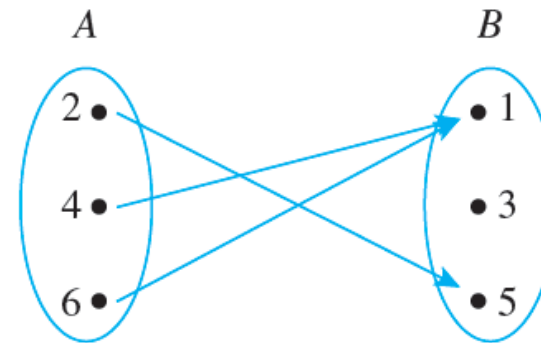
Example 1.3.4 – *Functions and Relations on Finite Sets*

Let $A = \{2, 4, 6\}$ and $B = \{1, 3, 5\}$. Which of the relations R , S , and T defined below are functions from A to B ?

a. $R = \{(2, 5), (4, 1), (4, 3), (6, 5)\}$

b. For every $(x, y) \in A \times B$, $(x, y) \in S$ means that $y = x + 1$.

c. T is defined by the arrow diagram



Equality of Functions

A relation is a subset of a Cartesian product and a function is a special kind of relation.

Specifically, if f and g are functions from a set A to a set B , then

$$f = \{(x, y) \in A \times B \mid y = f(x)\} \quad \text{and} \quad g = \{(x, y) \in A \times B \mid y = g(x)\}.$$

It follows that

f equals g , written $f = g$, if, and only if, $f(x) = g(x)$ for all x in A .

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Example of Graph

The word “graph” has different meanings. In the context of discrete mathematics (or even computer science), **graph** refers to a special mathematical structure.

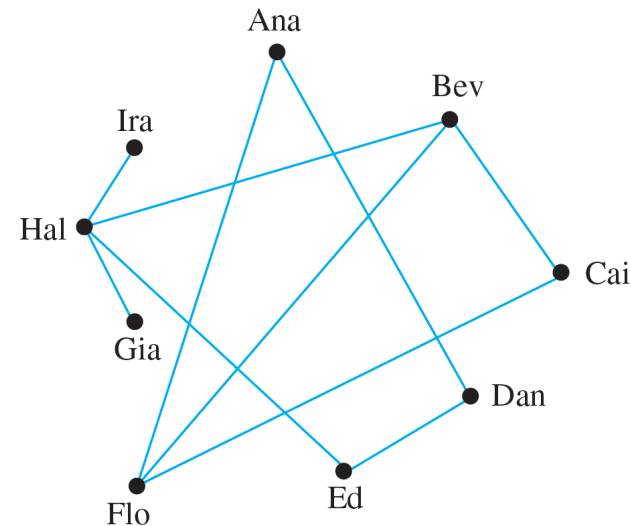
Imagine an organization that wants to set up **teams of three** to work on some projects.

In order to maximize the number of people on each team who had previous experience working together successfully, the director asked the members to provide names of their previous partners.

Example of Graph

This information is displayed below both in a table and in a diagram.

Name	Previous Partners
Ana	Dan, Flo
Bev	Cai, Flo, Hal
Cai	Bev, Flo
Dan	Ana, Ed
Ed	Dan, Hal
Flo	Cai, Bev, Ana
Gia	Hal
Hal	Gia, Ed, Bev, Ira
Ira	Hal

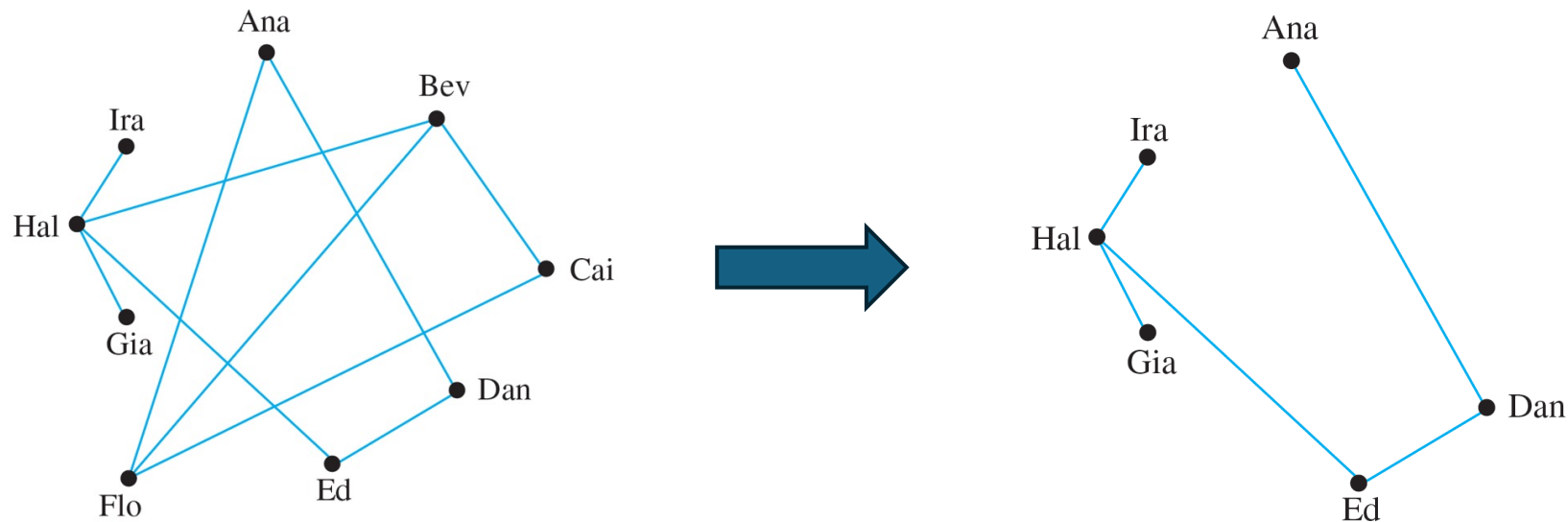


Can you find groups of three who collaborated before?

Example of Graph

It is easy to see that **Bev, Cai, and Flo** are a group of three previous partners, and so it would be reasonable for them to form one of these teams.

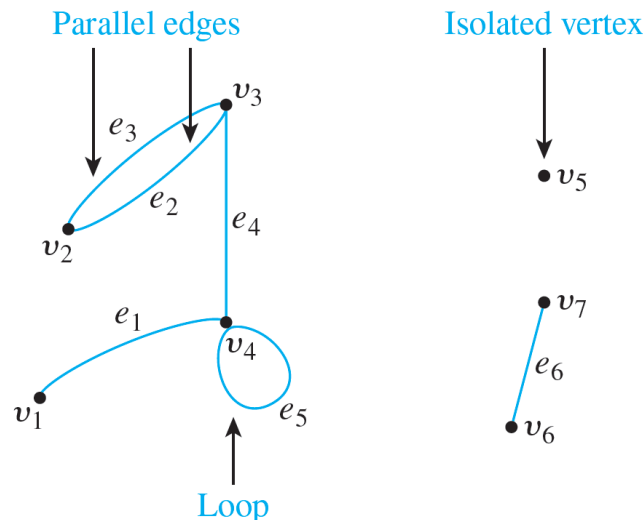
The drawing below shows the result when these three names are removed from the diagram.



The Language of Graphs

Drawings such as these are illustrations of a structure known as a **graph**. The dots are called **vertices** (plural of **vertex**) and the line segments joining vertices are called **edges**.

In general, a graph consists of **a set of vertices** and **a set of edges** connecting various pairs of vertices. The edges may be straight or curved and should either connect one vertex to another or a vertex to itself, as shown below.



The Language of Graphs

In this drawing, the vertices are labeled with v 's and the edges with e 's. When an edge connects a vertex to itself (as e_5 does), it is called a **loop**.

When two edges connect the same pair of vertices (as e_2 and e_3 do), they are said to be **parallel**. It is quite possible for a vertex to be unconnected by an edge to any other vertex in the graph (as v_5 is), and in that case the vertex is said to be **isolated**.

Formal Definition of a Graph

Definition

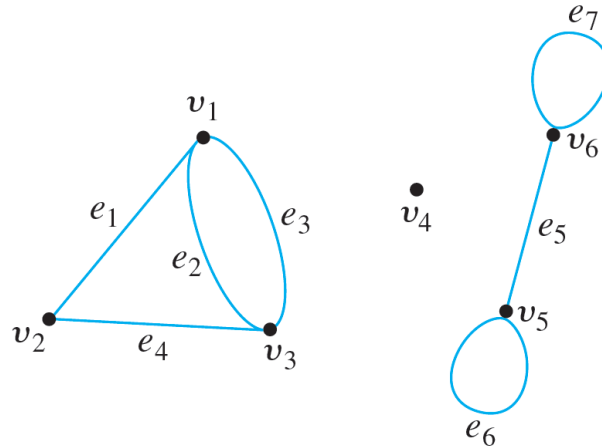
A **graph** G consists of two finite sets: a nonempty set $V(G)$ of **vertices** and a set $E(G)$ of **edges**, where each edge is associated with a set consisting of either one or two vertices called its **endpoints**. The correspondence from edges to endpoints is called the **edge-endpoint function**.

An edge with just one endpoint is called a **loop**, and two or more distinct edges with the same set of endpoints are said to be **parallel**. An edge is said to **connect** its endpoints; two vertices that are connected by an edge are called **adjacent**; and a vertex that is an endpoint of a loop is said to be **adjacent to itself**.

An edge is said to be **incident on** each of its endpoints, and two edges incident on the same endpoint are called **adjacent**. A vertex on which no edges are incident is called **isolated**.

Example 1.4.1 – *Terminology*

Consider the following graph:



- Write the vertex set and the edge set, and give a table showing the edge-endpoint function.
- Find all edges that are incident on v_1 , all vertices that are adjacent to v_1 , all edges that are adjacent to e_1 , all loops, all parallel edges, all vertices that are adjacent to themselves, and all isolated vertices.

Example 1.4.2 – *Drawing More Than One Picture for a Graph*

Consider the graph specified as follows:

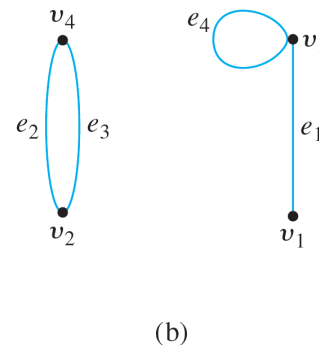
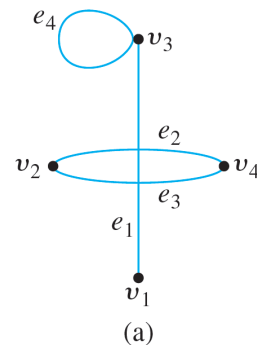
vertex set = $\{v_1, v_2, v_3, v_4\}$

edge set = $\{e_1, e_2, e_3, e_4\}$

edge-endpoint function:

Edge	Endpoints
e_1	$\{v_1, v_3\}$
e_2	$\{v_2, v_4\}$
e_3	$\{v_2, v_4\}$
e_4	$\{v_3\}$

Both drawings (a) and (b) shown below are pictorial representations of this graph.



Example 1.4.3 – *Labeling Drawings to Show They Represent the Same Graph*

Consider the two drawings shown in Figure 1.4.1. Label vertices and edges in such a way that both drawings represent the same graph.

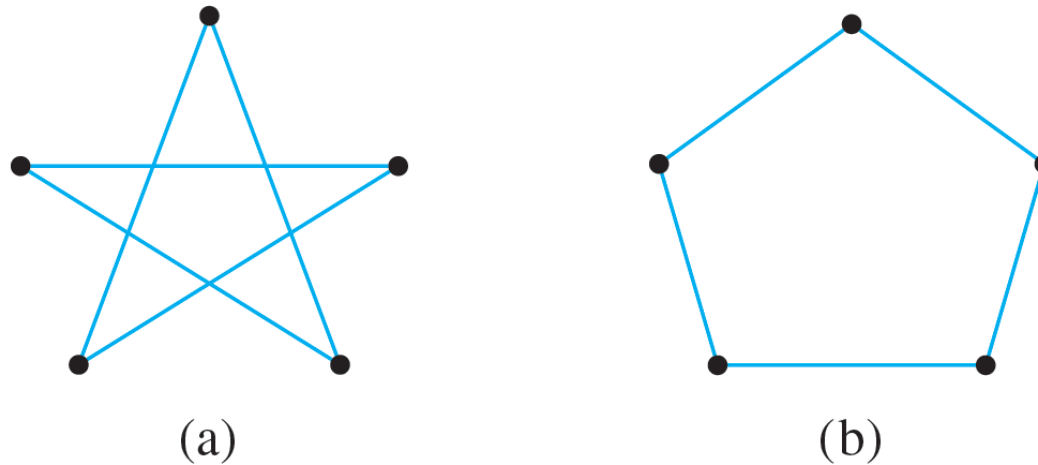
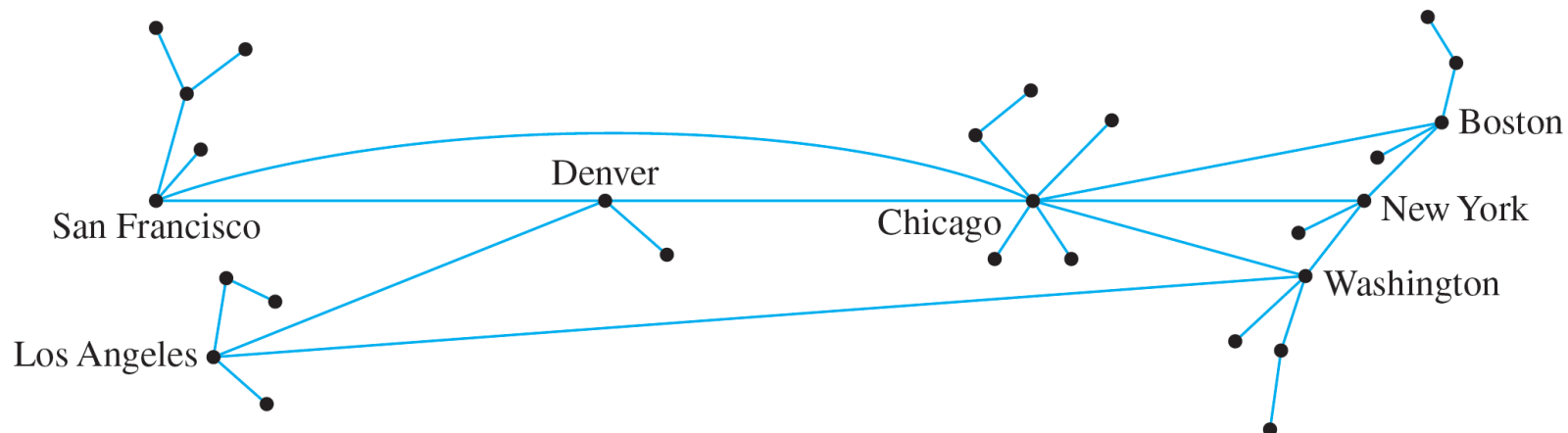


Figure 1.4.1

Example 1.4.4 – *Using a Graph to Represent a Network*

Telephone, electric power, gas pipeline, and air transport systems can all be represented by graphs, as can computer networks—from small **local area networks** to the **global Internet** system that connects millions of computers worldwide.

Questions that arise in the design of such systems involve choosing connecting edges to minimize cost, optimize a certain type of service, and so forth.



Directed Graph

A **directed graph** is like an (undirected) graph except that each edge is associated with an *ordered pair* of vertices rather than a *set* of vertices. Thus each edge of a directed graph can be drawn as an arrow going from the first vertex to the second vertex of the ordered pair.

Definition

A **directed graph**, or **digraph**, consists of two finite sets: a nonempty set $V(G)$ of vertices and a set $D(G)$ of directed edges, where each is associated with an ordered pair of vertices called its **endpoints**. If edge e is associated with the pair (v, w) of vertices, then e is said to be the **(directed) edge** from v to w .

Example 1.4.6 – *Knowledge Graph: Using a Graph to Represent Knowledge*

In many applications of artificial intelligence, a knowledge base of information is collected and represented inside a computer.

Because of the way the knowledge is represented and because of the properties that govern the artificial intelligence program, the computer is not limited to retrieving data in the same form as it was entered; it can also **derive new facts from the knowledge base** by using certain built-in rules of inference.

For example, from the knowledge that **the Los Angeles Times is a big-city daily** and that **a big-city daily contains national news**, an artificial intelligence program could **infer** that the **Los Angeles Times contains national news**.

Example 1.4.6 – Using a Graph to Represent Knowledge

The directed graph shown in Figure 1.4.2 is a pictorial representation for a simplified knowledge base about periodical publications.

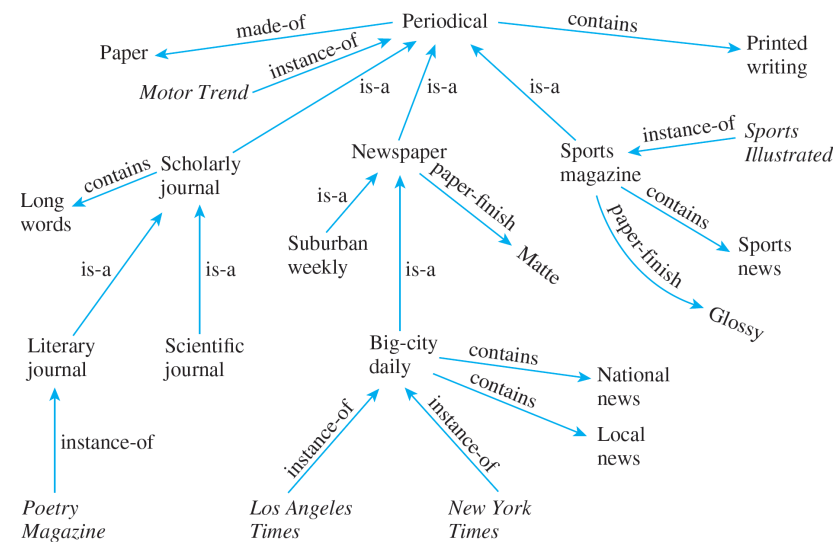


Figure 1.4.2

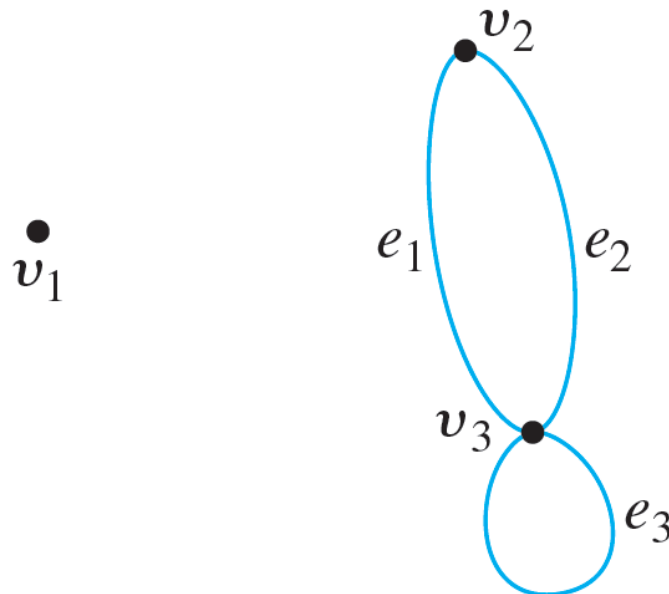
Question: According to this knowledge base, what paper finish does the *New York Times* use?

Vertex Degree

Definition

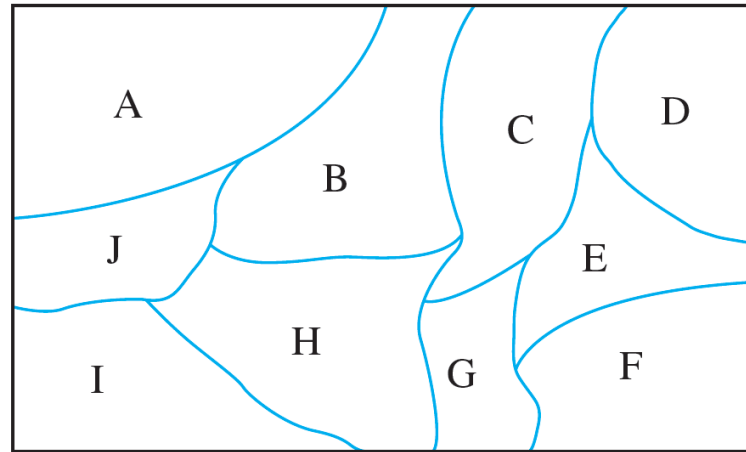
Let G be a graph and v a vertex of G . The **degree of v** , denoted $\deg(v)$, equals the number of edges that are incident on v , with an edge that is a loop counted twice.

Find the degree of each vertex of the graph G shown below.



Example 1.4.9 – *Using a Graph to Color a Map*

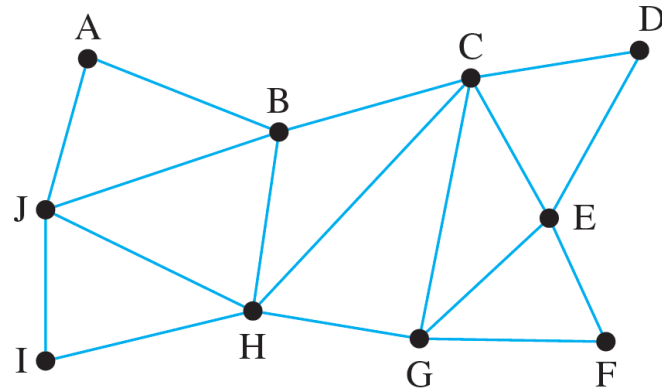
Imagine that the diagram shown below is a map with countries labeled *A–J*. Show that you can color the map so that no two adjacent countries have the same color.



Example 1.4.9 – *Solution*

Notice that coloring the map does not depend on the sizes or shapes of the countries, but only on which countries are adjacent to which.

So, to figure out a coloring, you can draw a graph, as shown below, where **vertices represent countries** and where **edges** are drawn between pairs of vertices that represent **adjacent countries**.



Example 1.4.9 – *Solution*

continued

Coloring the vertices of the graph will translate to coloring the countries on the map.

As you assign colors to vertices, a relatively efficient strategy is, at each stage, **to focus on an uncolored vertex that has maximum degree**, in other words that is connected to a maximum number of other uncolored vertices.

If there is more than one such vertex, it does not matter which you choose because there are often several acceptable colorings for a given graph.

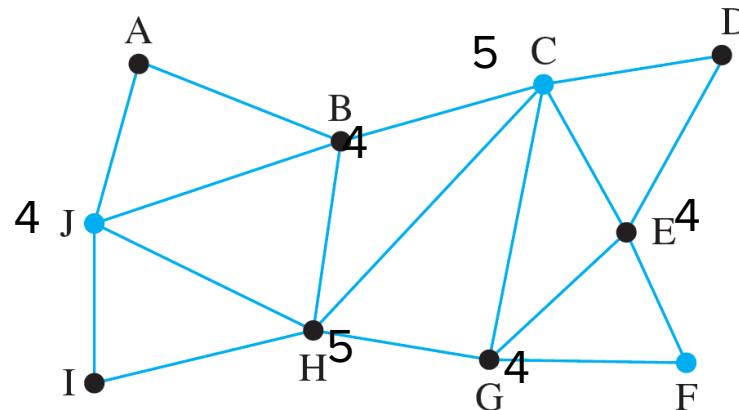
Example 1.4.9 – Solution

continued

For this graph, both C and H have maximum degree so you can choose one, say, C , and color it, say, blue.

Now since A , F , I , and J are not connected to C , some of them may also be colored blue, and, because J is connected to a maximum number of others, you could start by coloring it blue.

Then F is the only remaining vertex not connected to either C or J , so you can also color F blue. The drawing below shows the graph with vertices C , J , and F colored blue.



Example 1.4.9 – *Solution*

continued

Since the vertices adjacent to C , J , and F cannot be colored blue, you can simplify the job of choosing additional colors by removing C , J , and F and the edges connecting them to adjacent vertices. The result is shown in Figure 1.4.4a.

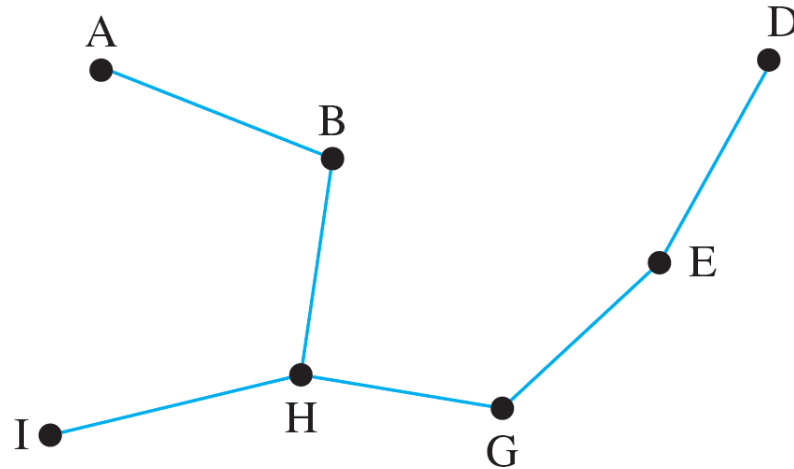


Figure 1.4.4(a)

Example 1.4.9 – Solution

continued

In the simplified graph again choose a vertex that has a maximum degree, namely H , and give it a second color, say, gray.

Since A , D , and E are not connected to H , some of them may also be colored gray, and, because E is connected to a maximum number of these vertices, you could start by coloring E gray.

Then A is not connected to E , and so you can also color A gray. This is shown in Figure 1.4.4b.

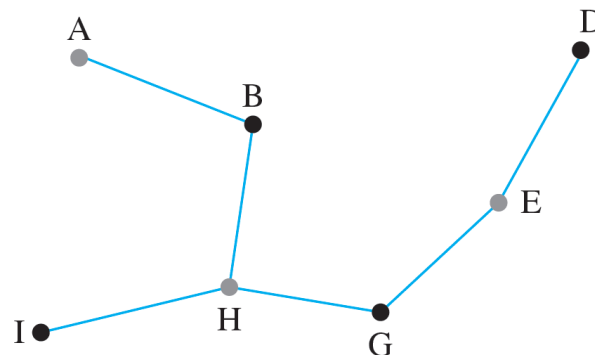
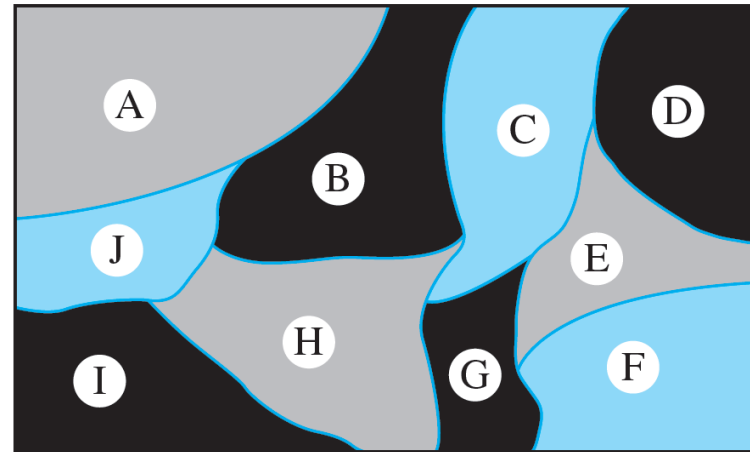
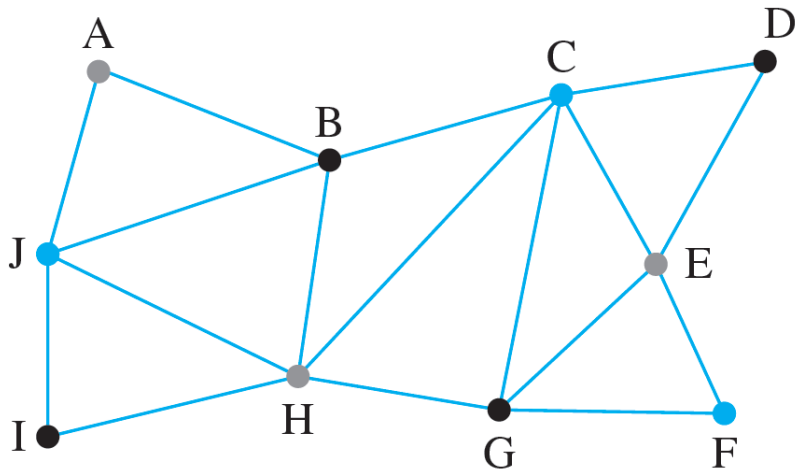


Figure 1.4.4(b)

Example 1.4.9 – *Solution*

continued

The drawing below shows the original graph with vertices C , J , and F colored blue, vertices H , A , and E , colored gray, and the remaining vertices colored black. You can check that no two adjacent vertices have the same color.



Four-Color Theorem

- **Four color theorem**: no more than four colors are required to color the regions of any map so that no two adjacent regions have the same color.
 - Remark: Adjacent means that two regions share a common boundary of non-zero length.
- Discovered in 1852 as **four color conjecture**.
- Many mathematicians tried to prove it but failed.
- In 1976, Kenneth Appel and Wolfgang Haken provide it with the help of computer which examined almost 2000 special cases.